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Programmable genome editing in human cells using RNA-guided bridge recombinases

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Site-specific insertion of gene-sized DNA fragments remains an unmet need in the genome editing field. IS110-family serine recombinases have recently been shown to mediate programmable DNA recombination in bacteria using a bispecific RNA guide (bridge RNA) that simultaneously recognizes target and donor sites. Here, we show that the bridge recombinase ISCro4 is highly active in human cells, and provide structural insights into its enhanced activity. Using plasmid- or all-RNA-based delivery, ISCro4 supports programmable multi-kilobase exisions and inversions, and facilitates donor DNA insertion at genomic sites with efficiencies exceeding 6%. Finally, we assess ISCro4 specificity and off-target activity. These results establish a framework for the development of bridge recombinases as next-generation tools for editing modalities that are beyond the capabilities of current technologies.

CRISPR-Cas genome editing technologies enable targeted gene disruption or correction of mutations at endogenous genomic loci using programmable guide RNAs (*1*). First-generation technologies, based on CRISPR-associated nucleases Cas9 and Cas12a, can efficiently induce gene knockouts (*2*–*7*). However, these approaches rely on the generation of DNA double-strand breaks (DSBs), which typically result in heterogeneous editing outcomes due to imprecise end-repair mechanisms and are associated with cellular genotoxicity (*8*–*10*). Moreover, their ability to mediate precise gene correction is limited by the dependence on endogenous homology-directed repair (HDR) pathways, which are largely inactive in postmitotic cells (*1*). To overcome these limitations, second-generation technologies, namely base and prime editors, have been developed to avoid the need for DSBs and HDR (*11*–*13*). Base editors enable programmable single-nucleotide substitutions (*11*, *12*), while prime editors can introduce more extensive modifications spanning several tens of nucleotides (*13*).

CRISPR-Cas genome editing has already revolutionized the treatment of genetic disorders, (*14*, *15*). However, many genetic diseases, including cystic fibrosis (*16*) and retinitis pigmentosa (*17*), are caused by a variety of mutations within the same gene. Effective treatment of such multiallelic genetic disorders necessitates corrective strategies based on site-specific insertion of large, gene-sized DNA payloads, which is challenging to achieve using existing first- and second-generation CRISPR genome editors. Several emerging technologies have aimed to address this unmet need, including prime-editing-assisted site-specific integrase gene editing (PASSIGE) (*18*–*24*), CRISPR-associated transposons (CASTs) (*25*–*27*), engineered retrotransposons (*28*), as well as fusions of catalytically inactive Cas9 with transposases (*29*, *30*). However, these approaches suffer from several limitations, including the formation of intermediate editing products that can disrupt gene function (*18*–*24*), large coding sizes that may hinder efficient cellular delivery (*18*–*28*), off-target insertions (*29*, *30*), and the creation of genomic scars, including recombinase landing pads (*18*–*24*) or transposon ends (*25*–*27*). In this context, the recently discovered bridge RNA-guided recombinases (*31*–*34*), originating from the IS110 family of bacterial insertion sequence (IS) transposable elements, represent a promising class of programmable systems for genome engineering applications.

IS110 bridge recombinases have been shown to catalyze site-specific recombination between target and donor DNAs using a bipartite RNA guide (termed bridge RNA or seekRNA) that mediates recognition of 14-nucleotide target and donor sites defined by the presence of an invariant "core" sequence, a CT dinucleotide in the case of the prototypical bridge recombinase IS621. The bridge RNA comprises two internal loops: a target-binding loop (TBL) and a donor-binding loop (DBL), each of which contains two variable segments that base-pair with the top and bottom strands of the target and donor sites (Fig. 1A) (*32*–*34*). Structural and biochemical studies of the IS621 recombinase have revealed that the recombination mechanism involves initial nicking of the target and donor top strands, followed by strand exchange and ligation to generate a Holliday junction intermediate, and finally junction resolution by bottom strand cleavage. The recombinase reaction is facilitated by the formation of a tetrameric synaptic complex composed of TBL- and DBL-bound recombinase dimers, which leads to the formation of composite active sites comprising a DEDD RuvC-like motif and a conserved catalytic serine residue. The tetrameric architecture of the bridge recombinase synaptic complex reflects structural homology with other serine recombinases and integrases, and distant evolutionary relationships with other non-programmable recombinase families, including Cre (*33*).

In their native setting, bridge recombinases function in conjunction with their RNA guides to mediate the mobilization of the insertion sequence elements that encode them. Previous studies have demonstrated that the bridge RNA guide can be reprogrammed to direct the IS621 recombinase to catalyze precise DNA rearrangements, including deletions, inversions, and insertions (Fig. 1A), both in vitro and in bacteria (*32*). Given their tremendous translational potential, we sought to test whether bridge recombinases can be repurposed for mammalian genome editing. We report that the ISCro4 recombinase from *Citrobacter rodentium* (*35*) supports high levels of RNA-guided recombination in cultured human-derived cells and provide structural insights into its activity based on a cryogenic electron microscopy (cryo-EM) reconstruction of the ISCro4 synaptic complex. We show that the activity of ISCro4 is further enhanced by splitting the bridge guide RNA into independent TBL and DBL guides, either expressed from plasmids or delivered as synthetic, chemically modified RNAs along with ISCro4 mRNA. The split-bridge guide RNAs can be robustly programmed to direct ISCro4-mediated chromosomal DNA rearrangements, supporting recombination frequencies of more than 10% for deletions and inversions in genome-integrated reporter cell lines, and over 6% for the insertion of exogenous multi-kilobase DNA fragments at endogenous loci. As proof of concept, we demonstrate corrective editing at several disease-relevant genomic loci, and provide insights into the specificity and off-target activity of ISCro4. Collectively, these results establish ISCro4 as a versatile molecular system for programmable genome rearrangements in mammalian cells.

Results

ISCro4 recombinase has high activity in mammalian cells

Inspired by the initial discovery and characterization of bridge recombinases (*31*), we sought to implement them for mammalian genome editing. The IS621 recombinase was previously shown to mediate a broad range of edits, including deletions, inversions and, insertions, in bacterial cells (*32*). Using the human-derived HEK293T cell line, we initially aimed to demonstrate recombination-mediated deletion in a reporter plasmid containing a protein-coding gene cassette flanked by native donor and target sequences from the IS621 element (*32*). Cells were co-transfected with the reporter plasmid, together with plasmids encoding IS621 recombinase and its native bridge RNA. The bridge recombinase–encoding sequence was codon-optimized for mammalian cell expression, linked to an N-terminal SV40 and a C-terminal nucleoplasmin nuclear localization sequence (NLS), and expressed under the control of a mammalian cytomegalovirus (CMV) promoter. Expression of the full-length bridge RNA was driven by a U6 RNA polymerase III promoter (Fig. 1C). Using a nested PCR assay, we were able to detect deletions resulting from recombination between the donor and target sites (fig. S1A). Recombination was dependent on bridge RNA expression and the catalytic activity of IS621 recombinase, as inactivating mutations in the recombinase active site (D11A or S241A) resulted in loss of recombination. The accuracy of recombination was confirmed by Sanger sequencing (fig. S1B). To quantify recombination efficiency, we used a modified Stoplight reporter system, initially developed for the Cre recombinase (*36*), in which productive recombination results in the deletion of a monomeric infrared fluorescent protein (mIFP) expression cassette and concomitant switch to the expression of the mCerulean fluorescent protein (fig. S1C). Using flow cytometry analysis, we were unable to detect an increase in mCerulean expression in HEK293T cells co-expressing the IS621 recombinase and its native bridge RNA, as compared to control cells expressing IS621 with a bridge RNA containing scrambled target and donor loop sequences. However, expression of a modified guide RNA design incorporating handshake base-pairing and an extended right-target guide (RTG) segment (*33*) to facilitate top strand recognition and exchange resulted in detectable recombination in approximately 0.6% of cells (fig. S1, C and D). Taken together, these results indicate that although IS621 recombinase is active in HEK293T cells, its efficiency is low.

Seeking to identify bridge recombinases with enhanced activity for genome engineering in human cells, we expanded our analysis to screen other IS110-family bridge recombinases from both IS110 and IS1111 subgroups. To this end, we selected a panel of ten candidate systems (Fig. 1B), including ISPpu10, ISAar29, ISHne5, ISCARN28, ISAzs32, for which the corresponding bridge RNA sequences were previously predicted (*32*), as well as ISPa11, ISEc21, ISkpn4, ISPst6, that were previously validated biochemically (*34*). We also included ISCro4 from an IS110-family system found in *Citrobacter rodentium* (*35*, *37*), for which we were able to predict the bridge RNA sequence by sequence alignments with the IS621 ortholog (*32*). Notably, the target and donor-binding sequences in the native ISCro4 bridge RNA naturally adopt handshake base-pairing. Initial screening of the candidates using the PCR-based plasmid deletion assay indicated that ISAar29, ISHne5, ISCro4, ISAzs32 and ISPst6 recombinases, in conjunction with their native bridge RNAs, were active in

HEK293T cells (Fig. 1D). After confirming that the sequences of the recombination products matched the expected outcomes (fig. S2), we quantified the activity of the most promising candidates using the Stoplight fluorescent reporter assay. The ISCro4 ortholog displayed the highest level of bridge RNA-dependent recombinase activity, resulting in recombination-mediated deletion in more than 36% of transfected cells (Fig. 1E). Collectively, these results establish ISCro4 as a highly efficient enzyme capable of executing RNA-guided recombination in mammalian cells.

Split-bridge RNA design boosts activity of ISCro4

Recent structural studies of the IS621 bridge recombinase have revealed that the enzyme adopts a tetrameric architecture that functions in conjunction with the TBL and DBL of the bridge RNA (*33*). Notably, the structure of the IS621 synaptic complex reveals that the linker connecting the TBL and DBL segments of the bridge RNA is not resolved, suggesting that it does not make intermolecular contacts with the recombinase. Furthemore, the observed interatomic distance between the 3′ end of the TBL and the 5′ end of the DBL suggests that the tetrameric synaptic complex assembles with two bridge RNA molecules (*33*). In light of these observations, we set out to test whether the TBL and DBL components of the bridge RNA could function *in trans*. We designed a split-bridge RNA format in which bridge RNA nucleotides 1–102 (comprising the TBL) and 110–179 (comprising the DBL) were provided as separate guide RNA molecules (Fig. 1F), and the linker sequence was omitted so as not to compromise TBL/DBL stability by exoribonuclease degradation. The functionality of the split TBL and DBL guides was initially validated using an in vitro recombination assay (fig. S3, A to F). In HEK293T cells, co-expression of TBL and DBL RNAs encoded by separate plasmids substantially improved editing efficiency (Fig. 1G and fig. S4, A and B), increasing the percentage of recombination-positive cells to 68% (Fig. 1G). The activity was dependent on co-expression of the RNAs, as expression of neither TBL nor DBL alone resulted in recombination. Furthermore, expression of ISCro4 catalytic mutants E60Q and S241A did not support split-bridge RNA-guided recombination, confirming that the observed activity was dependent on the catalytic activity of ISCro4 (Fig. 1G and fig. S4A).

Structural insights into the enhanced activity of ISCro4

Given the high degree of sequence identity between IS621 and ISCro4 (88%) recombinases (fig. S5A) and their respective native bridge RNA scaffolds (87%), we also investigated whether the split-bridge RNA design could improve the recombinase activity of IS621. The activity of IS621 recombinase co-expressed with separate IS621 TBL and DBL RNAs remained low, resulting in ~0.8% recombination-positive transfected cells. The fraction of recombination-positive transfected cells increased to ~2.1% of cells when the IS621 recombinase was co-expressed with the ISCro4 split-bridge RNAs (fig. S5B). In contrast, ISCro4, which was expressed at similar levels as IS621, was highly active irrespective of whether it was programmed with the ISCro4 or the IS621 split-bridge RNAs (fig. S5, B and C). Altogether, these observations suggest that the difference in recombination efficiencies of the IS621 and ISCro4 systems in mammalian cells is primarily due to the intrinsic activities of their respective recombinases, underpinned by their distinct amino acid sequences.

To obtain structural insights into the activity of ISCro4, we determined a cryo-EM reconstruction, at a resolution of 2.8 Å, of a synaptic complex comprising four ISCro4 protomers assembled together with DBL and TBL RNAs and their respective donor (dDNA) and target (tDNA) DNAs (Fig. 2 and figs. S6 and S7). The overall architecture of the complex is highly similar to that of the IS621 synaptic complex (*33*), with individual ISCro4 protomers superimposing with IS621 with a root-mean-square deviation of 0.86 Å (fig. S8A). The structure captures ISCro4 in state prior to top strand nicking and formation of covalent 5′-phosphoserine adducts between the donor/target top strands and the active site Ser241 (Fig. 2A). Of the 39 residues that differ between the amino acid sequences of ISCro4 and IS621, 7 residues are found in the vicinity of bound nucleic acids (Fig. 2B and fig. S5A), while 13 are found in the hydrophobic core and 19 are solvent-exposed without apparent nucleic acid contacts. Notably, Ser13$^{\mathrm{ISCro4}}$ (replacing Ala13$^{\mathrm{IS621}}$) makes an additional hydrogen bonding interaction with the backbone and nucleobases of the guide RNAs, respectively. Ser63$^{\mathrm{ISCro4}}$ (replacing Gly63$^{\mathrm{IS621}}$) makes hydrogen-bonding interaction with the nucleobases A81 (TBL) and A166 (DBL), involved in base pairing with the T nucleobases of the tDNA and dDNA CT cores (Fig. 2B and figs. S5A and S7), which likely preserves the CT core requirement in ISCro4. Ala15$^{\mathrm{ISCro4}}$ replaces Glu15$^{\mathrm{IS621}}$, removing a negative charge from the DNA binding interface, while Ser140$^{\mathrm{ISCro4}}$ (replacing Ala140$^{\mathrm{IS621}}$) is positioned close to TBL/DBL RNA backbone. Finally, Asn246$^{\mathrm{ISCro4}}$ and Arg247$^{\mathrm{ISCro4}}$ (replacing Arg246/Gly247$^{\mathrm{IS621}}$) perturb the electrostatic interactions within the Ser241 catalytic loop (Fig. 2B and figs. S5A and S7). Individual substitutions of these residues with corresponding ones from IS621 did not substantially impact ISCro4 recombination activity in the plasmid deletion reporter assay but their collective replacement resulted in a 21% reduction in the number of recombination-positive cells (fig. S8, B and C). Conversely, collective substitution of the corresponding residues in IS621 with their ISCro4 counterparts resulted in a ~2.4-fold increase in IS621 activity (fig. S8, B and C). Together, these observations indicate that the enhanced activity of ISCro4 is in part due to enhanced interactions with bound nucleic acids but mostly results from substitutions in the hydrophobic core and solvent-exposed parts of the molecular surface, likely impacting the in cellulo solubility and stability of the enzyme.

ISCro4 mediates genomic deletions, inversions and insertions

Having established that ISCro4 can efficiently delete gene-sized fragments from plasmids in HEK293T cells, we tested whether the bridge recombinase could also mediate large-scale inversions and insertions. To this end, we used redesigned reporter plasmids relying on the activation of mCerulean (mCer) protein expression as a signal of productive recombination (Fig. 3A). For the inversion reporter, the mCerulean-encoding cassette was placed in a reverse orientation with respect to the EF1α (elongation factor 1-alpha) promoter and flanked by the native ISCro4 donor and target sites. For insertion, we generated a target plasmid containing an EF1α promoter upstream of the ISCro4 target site, and used it in conjunction with a donor plasmid harboring a promoterless mCerulean gene downstream of an ISCro4 donor site. For HEK293T cells, recombination-dependent mCer expression was detected in 59%, 75% and 56% of transfected cells for the deletion, inversion and insertion plasmid reporters, respectively. For the same plasmid reporters in K562 cells, we observed deletion, inversion and insertion in 36%, 73% and 37% of transfected cells, respectively (Fig. 3B and fig. S9, A to C). We note that true "per-molecule" recombination efficiencies could not be quantified in these experiments due to the inability to determine the exact copy number of transfected reporter plasmids. Nevertheless, these results confirmed that ISCro4 is capable of mediating gene-size genetic rearragements in both HEK293T and K562 cell lines, suggesting that the approach is likely generalizable across mammalian cells.

Next, we tested whether ISCro4 could mediate RNA-guided recombination within chromosomal DNA. To this end, we randomly integrated the same recombination reporters in the genome of HEK293T cells using Tol2-mediated transgenesis (*38*). We modified the experimental protocol to utilize antibiotic selection for cells harboring the ISCro4 expression plasmid and assessed recombination three and seven days post transfection. Using this approach, recombination-dependent deletion and inversion was detected in approximately 10% and 15%, respectively, of puromycin-selected cells on day 3 (Fig. 3C and fig. S10, A and B). For ISCro4-dependent insertion, mCerulean expression was detected in approximately 2.3% of selected cells, as compared to 1.1% of control cells expressing a scrambled TBL guide (Fig. 3C and fig. S10, A and B). The precision of on-target recombination was validated by PCR assays (fig. S10C). For deletions and inversions, the fraction of recombination-positive cells remained largely stable over time. However, the fraction of insertion-positive cells was reduced by ~50% at the 7-day time point, suggesting that donor insertion may have impacted cell fitness (Fig. 3C).

To explore alternative delivery modalities, we used the same reporter cell lines and performed electroporation to deliver ISCro4-encoding mRNA together with synthetic TBL and DBL oligonucleotides containing protective 2'-O-methyl and phosphorothioate chemical modifications designed to enhance their stability (fig. S11A). This approach resulted in productive deletions and inversions in ~25% and ~10% of cells, respectively. For insertions, we were unable to detect productive recombination with a co-delivered donor plasmid, indicating the need for further optimization of the delivery protocol (Fig. 3D and fig. S11, B and C).

Reprogramming ISCro4 guide RNAs

We next sought to test whether the ISCro4 TBL and DBL RNA sequences could be selectively reprogrammed to direct recombination between new target and donor sequences. To this end, we used a reporter assay based on a universal target plasmid containing an EF1α promoter cassette upstream from the native ISCro4 target site and five additional CT-containing target sequences, in conjunction with a universal donor plasmid that contained an *EGFP* gene positioned downstream of the native ISCro4 donor site as well as five CT-containing sequences designed to serve as new donor sites (fig. S12A). The new target sites had Levenshtein sequence distances of 6–9 with respect to the ISCro4 native target sequence, while the donor sites had sequence distances of 4–10 relative to the ISCro4 native donor sequence. TBL and DBL RNAs were reprogrammed by varying TBL nucleotides 51-57 to base-pair with the bottom strand of the left-target (LT) sequence, and nucleotides 76-80 to pair with the top strand of the right-target (RT) sequence (fig. S7A). Similarly, DBL RNAs were designed by varying nucleotides 125-131 to pair with the bottom strand of the left donor (LD) sequence, while nucleotides 159-160 and 163-165 were designed to base-pair with the top strand of the right donor sequence (RD) (fig. S7B). Furthermore, the programming rules implemented handshake base pairing (*33*) to promote strand exchange during the recombination process by matching TBL nucleotides 83-84 with nucleotides 6-7 in the top donor strand and DBL nucleotides 168-169 with nucleotides 6-7 in the top target strand.

Monitoring recombination both using PCR analysis and flow cytometry, we observed that all of the reprogrammed TBL RNAs supported similar levels of on-target recombination (as compared to the ISCro4 native TBL RNA) when co-expressed with the native DBL RNA (fig. S12, B and C). Likewise, reprogrammed DBL RNAs were also functional in conjunction with native TBL RNA (fig. S12, D and E), although with slightly lower recombination efficiencies. Simultaneous TBL and DBL reprogramming also resulted in recombination at efficiencies comparable with those observed using native TBL and DBL guides (figs . S12, F and G, and S13A). On-target recombination between the reprogrammed target-donor pairs was also confirmed by Sanger sequencing (fig. S13, B and C). Taken together, these results demonstrate that the TBL and DBL RNA reprogramming rules are robust, enabling ISCro4 to be readily targeted to new donor and target sites.

ISCro4 enables editing at clinically relevant genomic loci

We proceeded with testing ISCro4 recombinase editing at several disease-relevant genomic loci in HEK293T cells. For deletion, we selected the *CNBP* (Cellular nucleic acid-binding protein) locus (Fig. 4A), as expansion of CCTG tetranucleotide repeats in the first intron of the *CNBP* gene is associated with type II myotonic dystrophy (*39*). We selected nine potential donor-target pairs that flank the *CNBP* CCTG repeats and tested whether reprogrammed TBL and DBL sequences could mediate excision of the repetitive sequences from the genome. PCR assays detected the expected deletions (299–535 bp) at the *CNBP* locus for four out of the nine TBL/DBL pairs tested (Fig. 4B). We selected two TBL/DBL pairs for follow-up analysis by next-generation sequencing (NGS), which confirmed that deletions were indeed caused by recombination between the chosen donor and target sites, with recombination efficiencies ranging from 0.5 to 1% (Fig. 4C and fig. S14A). In a second proof-of-concept deletion experiment, we aimed to remove the erythroid-specific enhancer from the *BCL11A* locus to mimic therapeutic genome editing strategies for sickle cell disease and beta-thalassemia (*15*). Out of four TBL/DBL guide RNA pairs tested, one combination was efficient at excising a 338-bp DNA sequence containing the *BCL11A* enhancer sequence (fig. S15A).

Next, we tested whether ISCro4 could mediate inversion of exon 23 of the *CFTR* (Cystic Fibrosis Transmembrane Conductance Regulator) gene (Fig. 4D), as the *CFTR*-W1282X mutation in exon 23 causes a severe cystic fibrosis form, and skipping exon 23 represents a potential corrective strategy (*40*). Five TBL/DBL guide combinations designed to recognize donor and target sites flanking exon 23 (out of nine tested) supported recombination, with the best performing TBL/DBL pair achieving inversion efficiency of approximately 4% (Fig. 4, E and F and fig. S14, B and C).

Finally, we used ISCro4 to integrate a 3.5-kb DNA fragment into the *AAVS1* (adeno-associated virus site 1) safe-harbor locus (Fig. 4G). To this end, we selected nine potential target sites at this locus and designed matching TBL guides, while maintaining the WT ISCro4 donor sequence in the donor plasmid. PCR assays showed that six out of the nine reprogrammed TBLs supported on-target donor insertions (Fig. 4H), and subsequent quantitative NGS analysis revealed insertion efficiencies ranging between 4% and 6% for the three best-performing TBL guides (Fig. 4I and fig. S14, D and E). Using the same TBL guides in K562 cells, we achieved insertion efficiencies of up to 0.4% (fig. S15B). Furthermore, we were also able to demonstrate donor insertion at the *CCR5* (chemokine C-C motif receptor 5) (*41*) locus in HEK293T cells (fig. S15C). Taken together, these results demonstrate the versatility of RNA-guided ISCro4 recombination in mediating a range of genome rearrangements at clinically relevant loci. Critically, we did not observe formation of ISCro4-dependent indels at the target genomic loci in our sequencing data, even though the catalytic mechanism of bridge recombinases involves sequential nicking of both the top and bottom strands in the donor and target. This suggests that the recombinase shields the DNA nicks during strand exchange and the recombination intermediates are thus not recognized as double-strand breaks by the end-joining DNA repair machinery.

Off-target activity of ISCro4

The questions of specificity and off-target activity are of critical importance for potential technological applications of bridge recombinases. To investigate these, we focused on the best-performing AAVS1 genomic insertion site (AAVS1 target #4). As the IS621 recombinase was previously shown to be capable of DBL-DBL recombination (*33*), we tested the cross-reactivity of ISCro4 guide RNAs using deletion plasmid reporters to quantify recombination programmed by DBL or TBL RNAs for the ISCro4 native donor and AAVS1 target sequences. Robust donor-to-target recombination could be achieved with both TBL-DBL combinations (TBL$^{\text{AAVS1}}$-DBL$^{\text{ISCRo4 donor}}$ or TBL$^{\text{ISCro4 donor}}$-DBL$^{\text{AAVS1}}$) (Fig. 5A and fig. S16A). For the AAVS1 target #4 sequence, no detectable TBL-TBL or DBL-DBL recombination was observed. In contrast, DBL-DBL recombination was substantial for the ISCro4 native donor sequence. For reference, no TBL-TBL or DBL-DBL recombination was detected for guides recognizing the native ISCro4 target sequence (fig. S16B), while substantial levels of both DBL-DBL and TBL-TBL recombination were observed for the native donor sequence in this experiment. These results indicate that ISCro4 can mediate DBL-DBL and TBL-TBL recombination at variable rates that are guide RNA-dependent.

We next examined the specificity of plasmid-to-genome integrations. Besides its intended site on chromosome 19, the AAVS1 target #4 sequence is found at five other loci in the human genome (table S1). We quantified insertion rates at the intended site as well as at four of these "perfect off-target" sites when recombination was programmed using the DBL$^{\text{ISCro4 donor}}$-TBL$^{\text{AAVS1}}$ (i.e., TBL recognizes the genome) or the

DBL^{AAVS1}-TBL$^{ISCro4\ donor}$ (i.e., DBL recognizes the genome) guide RNA pairs. We observed productive recombination at all sites, with frequencies ranging from 0.3% to 7.2% (Fig. 5B). Notably, targeting the genome with the DBL guide RNA resulted in slightly enhanced recombination for four out of five sites, but the magnitude of the effect was variable (Fig. 5B). These results suggest that efficiency of insertional recombination can be modulated by the choice of the guide RNA format programmed to recognize the genome and the donor DNA.

For initial assessment of the off-target activity of ISCro4, we tested the effect of single-nucleotide mismatches (transversion mutations) between the guides and their cognate substrates in the deletion plasmid reporter assay. We observed that both the TBL^{AAVS1} and DBL^{AAVS1} guides tolerated single mismatches at most positions within the 14-mer site, except for the CT core and the neighboring nucleotides at positions 10-11 in the right target segment (Fig. 5C and fig. S16C). For both TBL^{AAVS1} and DBL^{AAVS1}, recombination was entirely lost when the T nucleotide from the CT core was mismatched. These results indicate that ISCro4 recombination tolerates mismatches between the guides and their DNA substrates.

To investigate off-target DNA binding of ISCro4 and identify potential off-target recombination sites in cellulo, we first performed CUT&RUN analysis on HEK293T cells expressing FLAG-tagged ISCro4 and TBL or DBL guides targeting the AAVS1 #4 target site (fig. S17). We used a directed search strategy to profile candidate sites with 0–4 nucleotide mismatches to the guide RNA, filtered for CT core motif conservation. Using this approach, we identified off-target binding sites for both TBL^{AAVS1} (75 sites) and DBL^{AAVS1} (28 sites) guide RNAs that showed statistically significant level of enrichment over untransfected control cells. The aggregate mismatch profile of the sites reflected the single-mismatch data from plasmid reporter assays, with the majority of mismatches occurring at positions 1-7, while positions 10-14 were less variable. These results indicate that ISCro4 exhibits substantial off-target binding at mismatched genomic sites (fig. S17).

As the CUT&RUN analysis reflected DNA binding rather than recombination activity, we next assessed off-target donor insertion at a subset of the identified sites (Fig. 5D). We selected sites with high enrichment over background, as well as less-enriched sites containing one or two mismatches. We found that no detectable integration occurred at the most highly enriched CUT&RUN sites, all of which carried mismatches at position 6, which might disrupt handshake base pairing and thus selectively impact recombination. Insertion at other single- and double-mismatch sites occurred with frequencies ranging from 0% to 3%, as compared to the on-target insertion frequencies of roughly 2%. Using the DBL guide to target the genome generally resulted in lower off-target recombination levels, to a varying degree. Overall, these findings indicate that although ISCro4 exhibits considerable off-target binding in cellulo, its recombination activity appears to be limited to sites with up to two mismatches to the TBL or DBL guide. Furthermore, guide RNA orientation, i.e., whether the genome is recognized by the TBL or DBL, can selectively modulate off-target activity. However, the rules governing RNA-guided on- and off-target DNA recognition remain incompletely understood and further work will be required to elucidate the factors that affect recombination activity to inform optimal guide RNA design.

Addressing ISCro4 specificity constraints

Given the size of donor and target sites, the specificity of ISCro4 is inevitably constrained, since a 14-mer site is expected to be found on average in ~23 copies within the haploid human genome. Our genome insertion data for the AAVS1-targeting guides reveal detectable recombination at perfect off-target sites at comparable, albeit somewhat variable, frequencies that could reflect underlying local genome context, including chromatin accessibility or transcriptional status. To address this fundamental specificity constraint of bridge recombination, we performed targeted insertion experiments at unique sites within the human genome with exogenous DNA contaning donor sequences absent from the genome (i.e., genome-orthogonal donors). We selected four unique chomosomal sites (A–D), and tested whether each supported integration of a plasmid DNA carrying a genome-orthogonal donor site (1–4). Using corresponding TBL and DBL guide RNAs, ISCro4-mediated insertions at frequencies ranging from 0.4% to 2.8% (fig. S18A). In a plasmid reporter assay, no significant DBL-DBL recombination was detected for the four donor sites, further indicating that unintended DBL-DBL recombination is variable and depends on the underlying donor sequence (fig. S18B). Notably, chromosomal target site A is predicted to have no single-nucleotide mismatches elsewhere in the genome. We therefore examined whether donor integration occurred at a subset of predicted off-target sites containing two mismatches but detected no insertions (fig. S18C). These results suggest that stringent selection of unique genomic target sites based on their predicted mismatch profiles, combined with genome-orthogonal donors, could provide a feasible strategy to minimize off-target recombination activity for programmable genomic insertions (Fig. 5E).

Analysis of the hg38 reference human genome sequence reveals that of the 16,777,216 possible CT-containing 14-mers, 1,538,745 sites are found only once in the haploid human genome. 94% of all human protein-coding genes contain at least one unique site, and ~90% of all pathogenic gene variants listed in the ClinVar database contain at least one unique site in the first exon or first intron. Thus, the vast majority of the human protein-coding genome, including most disease gene variants, can in principle be targeted by a unique ISCro4 guide RNA for programmable DNA insertion. In turn, 1,758,019 possible genome-orthogonal 14-mers are available for designing guides to direct the insertion of exogenous DNA sequences. In the first line, these unique and genome-orthogonal site sets comprise the sequence space available for RNA-programmable genome manipulation in human cells using ISCro4.

Discussion

In this study, we demonstrate that the *Citrobacter rodentium* bridge RNA-guided recombinase ISCro4 exhibits robust activity in human-derived cells. By engineering the guide design to split the bridge RNA into standalone target- and donor-binding loops, and using refined RNA programming rules, we harness ISCro4 to install multi-kilobase insertions at a safe-harbor locus, as well as programmable deletions and inversions at disease-relevant loci. For deletions and inversions within genome-integrated reporter gene constructs, we report recombination rates exceeding 10%. In turn, programmable genomic insertion of exogenous DNA could be achieved with efficiencies exceeding 6%. Notably, we demonstrate chromosomal deletions of up to 1.6 kb, inversions of up to 800 bp, as well as targeted insertion of donor plasmids of up to 3.5 kb in size. Although we did not explicitly test the maximum size limits for the programmable edits, we expect ISCro4 to be capable of mediating megabase-scale genomic rearrangements, based on similarities with other site-specific recombinases such as Cre.

Our results are complementary to a recent study reporting successful implementation of ISCro4 for genome engineering in mammalian cells (*42*). The specificity of ISCro4 is inherently constrained by the 14-nt length of its donor and target sites, leading to detectable recombination at perfect and mismatched off-target sites in the genome. In the first line, this can be mitigated by selecting unique genomic target sites and relying on genome-orthogonal 14-mer sites in the donor DNA for insertions. Although unintended DBL-DBL (and potentially TBL-TBL) recombination is also of concern, the incidence of these off-target events appears to be sequence-dependent and can also likely be avoided by stringent selection of donor and target sites.We anticipate that ongoing molecular engineering of both ISCro4 and its guide RNA scaffolds will yield further activity and specificity enhancements. In this context, our structural analysis of ISCro4 reveals key molecular determinants contributing to its robust in cellulo activity, providing a framework to guide future rational engineering efforts.

Our work establishes ISCro4 as a promising system for editing modalities that remain challenging for CRISPR-based technologies and conventional, non-programmable recombinases and integrases. The recombinase mediates precise, scarless deletions and inversions without detectable indel formation, which may be applied for correction or modeling of pathogenic genome rearrangements. The ISCro4-mediated insertion mechanism bypasses the requirement for homology-directed repair (HDR) and could therefore support transgene insertion in HDR-deficient cells. Furthermore, unlike insertion technologies such as PASSIGE (*19*, *22*) or CRISPR-associated transposons (*25*–*27*), which also avoid DSB formation and HDR repair, ISCro4 inserts DNA with minimal genomic scarring, limited to duplication of the CT dinucleotide core. Together with compact size to facilitate cellular delivery, these properties position ISCro4, and bridge recombinases in general, as a versatile molecular platform to drive the development of novel genome engineering technologies for both basic research and therapeutic applications.

Materials and Methods

Phylogenetic analysis of bridge recombinases

The ISFinder database (*37*) was downloaded and insertion sequences (IS) corresponding to the IS110 family were selected (table S2). The 332 sequences were then converted to FASTA format and aligned using the MEGA software (*43*). Phylogenetic trees were constructed using the neighbor-joining method with 500 bootstrap replications. Finally, the trees were visualized in a circular format.

Ortholog selection and expression plasmid design

The screen included recombinases from insertion sequence (IS) elements for which bridge RNAs, as well as target and donor sequences, had been previously predicted (*32*, *34*). For ISCro4, the bridge RNA and target and donor sequences were identified by aligning the ISCro4 sequence with that of IS621. Although the ISFinder database provided only four nucleotides from the right target site of ISCro4, the extended RT sequence was inferred based on similarity to IS621 (*32*, *33*). Protein-coding sequences for all 11 IS systems were retrieved from the ISFinder database (*37*). Plasmids encoding bridge RNAs and recombinases were synthesized by Twist Bioscience and vector sequences are provided in table S1. Bridge RNAs were expressed under the control of the mammalian U6 promoter. To enhance expression by RNA polymerase III, each bridge RNA included a 5′-terminal guanine nucleotide (+G) and 3′-terminal olligo(T) sequence. Recombinase sequences were mammalian codon-optimized and expressed under the control of a CMV promoter. Each coding sequence was fused to a C-terminal 3×FLAG tag and an SV40 nuclear localization signal (NLS), as well as an N-terminal nucleoplasmin NLS.

Recombination reporter design

For the initial deletion-based nested PCR screens, a Cre Stoplight reporter (Addgene #45150) (*36*) was modified to monitor bridge recombinase activity. Specifically, the *loxH* site upstream of the dsRed gene was replaced with predicted wild-type donor sequences, and the *loxP* site downstream of dsRed was replaced with predicted wild-type target sequences. For flow cytometry assays, we adapted the reporter to express mIFP and mCerulean in place of dsRed and EGFP. The mIFP cassette was obtained from Addgene #200235, and mCerulean from Addgene #200249 (*44*). To reduce background expression of mCerulean in the absence of recombination, we inserted two microRNA-17 (miR-17) target sites downstream of the mIFP polyadenylation signal (*45*). To increase the system's versatility, the fluorescent reporter constructs were cloned into a plasmid backbone containing Tol2 arms (Addgene #200249 (*44*), allowing genomic integration when co-transfected with the Tol2 pHelper plasmid (Addgene #31823) (*38*).

The ISCro4 inversion reporter contains an mCerulean transgene positioned in the reverse orientation relative to the EF1α promoter. mCerulean is flanked by predicted wild-type donor and target sequences that are arranged in opposite orientations. In the absence of recombination, mCerulean is not expressed. Upon a positive recombination event, the mCerulean cassette is inverted to match the orientation of the promoter, enabling its expression. The reporter also includes a constitutively expressed mIFP transgene, which serves as a transfection control.

The ISCro4 insertion reporter contains an EF1α promoter and a wild-type target sequence. These reporters are co-transfected with donor plasmids encoding promoterless mCerulean or EGFP fluorophores. Wild-type donor sequences are placed upstream of the fluorophores to facilitate their integration under the control of the EF1α promoter following recombination. All relevant reporter construct and donor plasmid sequences are provided in table S1.

ISCro4 split bridge RNA design and TBL and DBL reprogramming rules

ISCro4 bridge RNA was divided into a target binding loop (TBL; positions 1–102) and a donor binding loop (DBL; positions 110–179) (*33*). TBL and DBL RNAs were reprogrammed by by varying TBL nucleotides 51-57 to base-pair with the bottom strand of the left-target (LT) sequence, and nucleotides 76-80 to pair with the top strand of the right-target (RT) sequence. DBL RNAs were designed by varying nucleotides 125-131 to pair with the bottom strand of the left donor (LD) sequence, and nucleotides 159-160 and 163-165 with the right donor sequence (RD). The programming incorporated handshake base by matching TBL nucleotides 83-84 with nucleotides 6-7 in the top donor strand and DBL nucleotides 168-169 with nucleotides 6-7 in the top target strand. Implementing these rules, TBL and DBL sequences for user-defined target-donor pairs were generated using a custom Python script (https://github.com/Schwank-Lab/ISCro4_Pelea_2026) (*46*). The script also generates overhang extension PCR primers for cloning reprogrammable TBL and DBL designs into the mammalian expression vector.

Molecular cloning of plasmid constructs

Fluorescent protein reporter and CMV-ISCro4-T2A-puromycin expression plasmids were assembled using the NEBuilder HiFi DNA Assembly Cloning Kit (NEB, E5520S) according to the manufacturer's instructions, with minor modifications. Reaction volumes were reduced to one-third of the recommended amounts. Incubations were carried out at 50°C for 2 hours for assemblies involving fewer than three fragments, and for 4 hours when assembling three or more fragments. PCR primers with 25–30 nucleotide overhangs were designed and ordered from Sigma-Aldrich. For the Stoplight deletion reporter plasmids, wild-type donor and target sequences were incorporated into the primer overhangs. PCR reactions were performed using Phusion High-Fidelity DNA Polymerase (Thermo Fisher, F530S) in 50 μl volumes, with an extension time of 1 min per kilobase.

PCR products were resolved on a 1% agarose gel, purified using the NucleoSpin Gel and PCR Clean-up Kit (Macherey-Nagel, 740610.20), and eluted in 44 μl of nuclease-free water. The eluted DNA was treated with 5 μl of Tango Buffer and 1 μl of DpnI (Thermo Fisher, ER1702), followed by incubation at 37°C for 1 hour. A second purification was performed using the same kit, with final elution in 15 μl of nuclease-free water. DNA concentrations were measured using a NanoDrop spectrophotometer, and fragments were combined at recommended molar ratios for assembly using the NEBuilder HiFi DNA Assembly Kit. Following assembly, 3 μl of the reaction was transformed into 15 μl of competent cells provided with the kit.

Plasmids for expression of split-bridge guide RNAs, reprogrammed TBL and DBL sequences, IS621 and ISCro4 catalytic mutants, and donor plasmids were cloned using overhang extension PCR. Primers were designed to anneal to the 5′ and 3′ ends of the plasmid backbone to be retained, incorporating 20–30 nucleotide overhangs. To minimize the risk of introducing unintended point mutations, primer lengths were kept under 60 nucleotides. PCR reactions were carried out in 50 μl volumes using Phusion High-Fidelity DNA Polymerase (Thermo Fisher, F530S), following the manufacturer's instructions, with an extension time of 1 min per kilobase. Following amplification, reactions were treated with 5.6 μl of Tango Buffer and 1 μl of DpnI (Thermo Fisher, ER1702), and incubated at 37°C for 1 hour to digest template DNA. PCR products were resolved on a 1% agarose gel, purified using the NucleoSpin Gel and PCR Clean-up Kit (Macherey-Nagel, 740610.20), and eluted in 15 μl of elution buffer. A total of 3 μl of purified DNA was transformed into 25 μl of home-made competent MACH1 cells (Thermo Fisher, C862003). In parallel, we also performed transformation of 1 μl purified DNA into 5 μl of competent *E. coli* supplied with the NEBuilder HiFi DNA Assembly Cloning Kit (NEB, E5520S).

For both cloning methods, single colonies were picked after overnight incubation, cultured in LB medium supplemented with the appropriate antibiotic, and plasmids were extracted using the GeneJET Plasmid Miniprep Kit (Thermo Fisher, 7002425) and verified by either full-plasmid sequencing or Sanger sequencing (Microsynth).

Maintenance of HEK293T and K562 cells

HEK293T and K562 cell lines were obtained from ATCC. HEK293T were cultured in full media containing DMEM (Thermo Fisher, 7001570) supplemented with 10% fetal bovine serum (FBS, Thermo Fisher, 7001624) and 1% Penicillin-Streptomycin (Gibco, 15070063). K562 cells (ATCC) were cultured in RPMI 1640 medium (Thermo Fisher Scientific, 11875093) supplemented with 10% FBS and 1% Penicillin–Streptomycin. Cells were cultured at 37°C and 5% CO_2. HEK293T cells were passaged every two days, whereas K562 cells were passaged every four to five days. Cell lines were routinely tested for mycoplasma contamination using Microsynth's mycoplasma testing service.

HEK293T cell transfection

For all HEK293T episomal plasmid assays, as well as plasmid DNA transfections of genome-integrated HEK293T reporter lines, transfections were performed using Polyethylenimine (PEI MAX, Polysciences, 24765-100). One day prior to transfection, HEK293T cells were seeded into 24-well plates at a density of 200,000 cells per well and cultured overnight in complete growth media. Thirty minutes before transfection, the medium was replaced with transfection media consisting of DMEM (Thermo Fisher, 7001570) supplemented with 2% fetal bovine serum (Thermo Fisher, 7001624). For each well, DNA mixtures were prepared in 50 μl Opti-MEM (Gibco, 7001587) using 3 μl of PEI for every 1 μg of DNA, incubated at room temperature for 20 min, and then added dropwise to the cells.

For plasmid-based assays, cells were returned to complete growth media one day after transfection and analyzed by flow cytometry or PCR-based assays 48 hours post-transfection. For chromosomal gene editing experiments, cells were switched to complete media containing 3 μg/ml puromycin one day after transfection. Selection was maintained for 48 hours to enrich for cells successfully transfected with the CMV-ISCro4-T2A-puromycin plasmid, followed by further analysis by flow cytometry or PCR. Samples analyzed 7 days post-transfection were moved back to puromycin-free medium at day 3 and maintained under normal culture conditions until analysis.

Each well was transfected with 500 ng of fluorescent reporter plasmid, 500 ng of recombinase expression plasmid, and 500 ng of bridge RNA-expressing plasmid. For experiments using split-bridge RNA designs, 250 ng each of TBL and DBL plasmids were transfected. For endogenous gene editing, 500 ng of recombinase plasmid was co-transfected with 250 ng each of TBL and DBL plasmids. In experiments involving integrations, 750 ng of donor plasmid was also included. Samples denoted as TBL– or DBL– were transfected with plasmids expressing scrambled TBL and DBL sequences, respectively.

All experiments involving editing at endogenous HEK293T loci were performed using the TurboFect transfection reagent (Thermo Scientific, R0531). One day before transfection, HEK293T cells were seeded into 24-well plates at a density of 50,000 cells per well and cultured overnight in complete growth media. For deletion and inversion assays, each well received 250 ng of ISCro4-T2A-puromycin plasmid, 125 ng of TBL plasmid, and 125 ng of DBL plasmid. For insertion assays, each well was transfected with 150 ng ISCro4-T2A-puromycin plasmid, 75 ng each of TBL and DBL plasmids, and 200 ng of donor plasmid. For each transfection, 50 μl of Opti-MEM was added to the DNA mix (tube A). Separately, 2 μl of TurboFect reagent was mixed with 50 μl of Opti-MEM (tube B) and briefly vortexed. Tube B was then added to tube A, and the mixture was incubated at room temperature for 20 min. The resulting transfection mixture was added dropwise to the cells. Puromycin selection (3 μg/ml) was initiated 24 hours post-transfection to enrich for successfully transfected cells, and editing efficiency was assessed 72 hours after transfection.

Reporter cell lines

To generate stable reporter cell lines, HEK293T cells were seeded in 6-well plates and transfected with PEI. Each well received 500 ng of reporter plasmid and 2000 ng of Tol2 pHelper plasmid (pCMV-Tol2; Addgene #31823) (*38*). To enrich for cells with stable genomic integrations, two rounds of fluorescence-activated cell sorting (FACS) were performed: the first one-week post-transfection and the second three weeks post-transfection, selecting for $mIFP^+$ cells. After the second round, cells were passaged at least twice before use in subsequent experiments. FACS sorting was carried out on a BD FACSymphony S6 5L instrument.

ISCro4 mRNA production

A linearized mRNA production plasmid for ISCro4 was used as a template for the in vitro transcription using the high-yield T7 P&L RNA synthesis kit from Jena Biosciences. Modified nucleoside-containing mRNA was generated using N1mΨ-5′-triphosphate (TriLink) instead of UTP. Co-transcriptional addition of the trinucleotide cap1 analog

CleanCap (TriLink) was used to cap the in vitro transcribed mRNAs. mRNA was purified using Monarch columns (NEB) and eluted in nuclease-free water. The resulting mRNAs were analyzed by agarose gel electrophoresis and stored at -80 °C until use.

mRNA electroporation of HEK293T cells

For electroporation experiments, an mRNA mix was freshly prepared on ice by combining 2 µg ISCro4 mRNA, 100 pmol synthetic TBL RNA, and 100 pmol synthetic DBL RNA containing chemical modifications (see table S1 and https://github.com/Schwank-Lab/ISCro4_Pelea_2026) (*46*). For insertion assays, 500 ng donor plasmid DNA was additionally included in the mix. HEK293T cells were trypsinized, diluted in complete medium, pelleted at 300 × g for 2 min, washed twice with PBS, and resuspended in supplemented SF Cell Line 4D-Nucleofector® X Solution (Lonza, V4XC-2032). A total of 500,000 cells were used per electroporation. Briefly, 20 µl of the resuspended cell suspension was added to the RNA (or RNA + donor DNA) mix, gently mixed, and electroporated using the Lonza 4D-Nucleofector® X Unit with the HEK293T program. Immediately after electroporation, 80 µl of pre-warmed DMEM was added to each cuvette, and cells were allowed to recover for 10 min at 37°C. The recovered cells were then transferred to 24-well plates pre-filled with 1 ml complete medium and incubated under standard culture conditions (37°C, 5% CO_2).

Plasmid DNA electroporation of K562 cells

K562 cells were nucleofected as previously described (*47*) using the Lonza 4D-Nucleofector® X Unit. Briefly, for the K562 plasmid-based deletion and inversion assays (Fig. 3), 330 ng ISCro4 expression plasmid, 330 ng reporter plasmid, 165 ng TBL plasmid, and 165 ng DBL plasmid were used per reaction. Nucleofection was carried out using a Lonza 4D-Nucleofector 16-well strip and the FF-120 program by mixing plasmid DNA with 500,000 cells (each well) in 20 µl of homemade nucleofection buffer (*48*). For insertion assays, an additional 500 ng donor plasmid was included. For AAVS1 genome editing experiments (fig. S14), each electroporation contained 300 ng ISCro4_T2A_puro plasmid, 400 ng donor plasmid, 150 ng TBL plasmid, and 150 ng DBL plasmid. K562 cells were selected with 1 µg/ml puromycin starting one day post-nucleofection to enrich for positively transfected cells and harvested three days post-nucleofection.

Nested PCR recombination assays

Cells were harvested for PCR assays by trypsinization, and genomic DNA was isolated using the Quick-gDNA MiniPrep kit (Zymogen, D3024), following the manufacturer's instructions. All PCR assays were performed using Phusion High-Fidelity DNA Polymerase (Thermo Fisher, F530S). For the initial ortholog screen, a nested PCR combined with restriction digestion was performed. The first PCR was carried out in a 20 µl reaction containing 8 µl of genomic DNA. This reaction ran for 11 cycles using primers flanking the dsRed gene in the fluorescent Stoplight recombination reporters, with a 1-min extension time. PCR products were purified using the NucleoSpin Gel and PCR Clean-up Kit and eluted in 20 µl of nuclease-free water. Next, 2 µl of the purified DNA was digested with a mixture of NEB restriction enzymes (BbsI, NotI, SbfI, and XbaI) that cut unique sites within the dsRed amplicon. These enzymes selectively digest non-recombined products while leaving recombined fragments intact. The digestion mix contained 2 µl DNA, 5 µl CutSmart buffer, 41 µl nuclease-free water, and 0.5 µl of each enzyme. Digestion was performed at 37°C for 1 hour, followed by enzyme inactivation at 65°C for 15 min. Finally, 2 µl of the digested DNA was used as template for a second PCR (35 cycles) in a 50 µl reaction with a 15-s extension time to enrich for smaller, digestion-resistant fragments. All remaining PCR-based assays using plasmid or genomic DNA were performed using nested PCR to reduce non-specific amplification from off-target primer binding. 4 µl of genomic DNA extract served as input for the first-round PCR in a 20 µl reaction with 11 cycles. The resulting PCR products were diluted 1:10 by adding 180 µl of nuclease-free water. These diluted products were then used as input for a 50 µl second-round PCR reaction. The second PCR included 30 cycles for plasmid-based assays and genome-integrated reporters, and 35 cycles for endogenous gene editing assays. PCR amplicons were resolved on a 1% agarose gel. Gel was prepared with SYBR Safe (Invitrogen, S33102) and 10 µl SYBR Safe were added per 150 ml agarose solution. Relevant PCR fragments were purified using the NucleoSpin Gel and PCR Clean-up Kit, followed by Sanger sequencing (Microsynth). A complete list of primer sequences is provided in table S1.

Sample preparation for flow cytometry analysis

Cells were first washed with phosphate-buffered saline (PBS, Gibco, 10010023), and the PBS was then aspirated. Cells were incubated in 100 µl Trypsin-EDTA (Gibco, 25200056) for 5 min at 37°C. Trypsin was inactivated by adding 500 µl of full media, and cells were transferred to 1.5 ml Eppendorf tubes. Samples were centrifuged at 300 × g for 5 min, the supernatant was removed, and cells were resuspended in 250 µl FACS buffer (PBS supplemented with 2% FBS). The cell suspension was filtered through a 70 µm cell strainer (Falcon, 352350) and transferred to round-bottom polystyrene test tubes (Falcon, 352058). Cells were kept on ice until analysis on a BD FACSymphony cytometer.

Data acquisition settings were as follows: forward scatter (FSC-A, FSC-H) voltage at 300, side scatter (SSC-A, SSC-H) voltage at 200. Fluorescent signals were detected using

Vio450/50 (voltage 350) for mCerulean, Red 670/30 (voltage 460) for mIFP, and Blue 530/30 (voltage 220) for EGFP.

Flow cytometry analysis

For HEK293T experiments, 100,000 events were collected per sample, whereas for K562 cell experiments, 50,000 events were acquired. Data analysis was performed using FlowJo (v10). HEK293T cells were identified by plotting FSC-A (x-axis) against SSC-A (y-axis). Single cells were then gated by plotting SSC-A (x-axis) against SSC-H (y-axis). Gates for the mIFP, mCerulean, and EGFP channels were established using a negative, non-transfected control sample.

For samples with fluorescent reporters where mIFP is constitutively expressed and mCerulean is activated upon positive recombination events, fluorescence data were analyzed by plotting mIFP on the x-axis and mCerulean on the y-axis. The percentage of mCerulean-positive transfected cells was calculated using the following formula:

$$\%mCer + \text{transfected cells} = \left[\frac{\left(\text{count of mCerulian}^{+}/\text{mIFP}^{-}\right)+\left(\text{count of mCerulian}^{+}/\text{mIFP}^{+}\right)}{\left(\text{count of mCerulian}^{+}/\text{mIFP}^{-}\right)+\left(\text{count of mCerulian}^{+}/\text{mIFP}^{+}\right)+\left(\text{count of mCerulian}^{-}/\text{mIFP}^{+}\right)}\right]\times 100$$

For samples where EGFP expression occurs upon recombination (fig. S12), the percentage of EGFP-positive cells was determined by plotting EGFP fluorescence (y-axis) against SSC-H (x-axis). To normalize $EGFP^{+}$ cell counts, activation with wild-type TBL and DBL sequences was set as 100%. mCerulean median fluorescence intensity (MFI) values were calculated for the cell population gated as positive for mIFP or mCerulean expression. Unless stating otherwise in the figure legends, bar graphs represent recombination activity measured across three biological replicates. Error bars indicate ± standard deviation, and individual data points from each replicate are shown as dots on the corresponding bars. Numerical values corresponding to each datapoint are provided in table S3.

Western blotting

Cells were seeded in 6-well plates and transfected with 1000 ng recombinase and 1000 ng bridge RNA plasmids using the PEI transfection protocol. Two days post-transfection, cells were washed with PBS, scraped, and pelleted by centrifugation at 300 × g for 5 min. Pellets were lysed in 500 µl lysis buffer (20 mM HEPES pH 7.5, 150 mM KCl, 0.2% Tween 20) and sonicated for 10 s (Bandelin Sonoplus, MS72 probe, 10% Amplitude). Lysates were cleared by centrifugation at 14,000 × g for 30 min at 4°C. A 300 µl aliquot was mixed with 60 µl 5× SDS sample buffer, heated at 95°C for 10 min, and loaded onto a precast Any kDa gel (Bio-Rad, 4569036). Protein transfer was performed using the Bio-Rad Trans Blot Turbo System following the manufacturer's instructions and using Trans-Blot Turbo Mini PVDF membranes (Bio-Rad 1704156). Membranes were blocked for 1 hour at room temperature with shaking in 1× TBST containing 4% milk powder (TBST: 20 mM Tris-HCl pH 7.5, 150 mM NaCl, 0.1% Tween 20). They were then incubated for 1 hour at room temperature with anti-FLAG (Sigma, A8592-2MG) or anti-GAPDH (Proteintech, 10494-1-AP) primary antibodies diluted 1:10,000 in 1× TBST with 4% milk. After washing three times for 15 min each with 1× TBST, FLAG detection was performed using Pierce ECL Western Blotting Substrate (Thermo, 32209) according to the manufacturer's instructions. For GAPDH, membranes were incubated with anti-rabbit secondary antibodies (Sigma, A9169-2ML) at 1:10,000 dilution for 1 hour, washed three times, and developed. Western blots were imaged using the Bio-Rad ChemiDoc system. To quantify the relative protein expression, raw image data were exported to .tiff format (Image Lab Version 6.1.0 build 7) and imported into FIJI. Bands for FLAG and GAPDH signal were selected manually and quantified volumetrically. Intensity levels were exported, and each FLAG signal was divided by its corresponding GAPDH signal.

ISCro4 protein expression and purification

ISCro4 recombinase, fused with N-terminal MBP tag and a C-terminal 6×His tag, was expressed in *E. coli* Rosetta2 (DE3) cells (Novagen, 71397). Cultures were grown at 37 °C with shaking at 110 rpm until reaching an OD_{600} of 0.6. Protein expression was induced with 0.5 mM IPTG and allowed to proceed for 4 hours at 37 °C. Harvested cells were resuspended in lysis buffer (20 mM Tris-HCl, pH 8.0; 1 M NaCl; 5 mM $MgCl_2$; 5 mM imidazole; and protease inhibitor cocktail) and lysed at 4 °C using a Maximator® high-pressure homogenizer (3 passes at 25,000 psi). The lysate was clarified by centrifugation at 44,000 × g for 25 min at 4 °C, and the supernatant was loaded onto a 5 ml Ni-NTA Superflow cartridge (Qiagen, 1034558). Protein was eluted using Ni-NTA elution buffer (20 mM Tris-HCl, pH 8.0; 500 mM NaCl; 5 mM $MgCl_2$; 140 mM imidazole). The MBP tag was removed by incubating the eluted protein with His-tagged TEV protease (1:50 molar ratio) for 4 hours at 4 °C. The cleavage reaction was loaded onto a 5 ml HiTrap Heparin HP cartridge (Cytiva, 7040703), and the protein was eluted using a 50 ml linear gradient of Heparin elution buffer (20 mM Tris-HCl, pH 8.0; 1 M NaCl; 5 mM $MgCl_2$). The eluted fractions were further purified by size-exclusion chromatography (SEC) on a Superdex 200 Increase 10/300 column (Cytiva, 28990944). Peak fractions were collected, concentrated to ~4 mg/ml, and stored at −80 °C until use. The E60Q and S241A mutant proteins were expressed and purified using the same protocol.

ISCro4 in vitro recombination assay

In vitro assays were carried out as described previously for IS621 (*33*). Cy5-labeled target DNA (tDNA) and donor DNA (dDNA) oligonucleotides were sourced from Integrated DNA Technologies (IDT), while unlabeled oligos were purchased from Sigma-Aldrich. Full-length bridge RNAs, TBL, and DBL were generated by in vitro transcription using a previously published protocol (*49*). Oligonucleotide and RNA sequences are listed in table S1. ISCro4 protein (40 µM) was preincubated on ice for 20 min in recombination buffer (20 mM Tris-HCl, pH 7.5; 300 mM NaCl; 5 mM $MgCl_2$, 1 mM DTT) with one of the following RNA concentrations: 10 µM full-length bridge RNA, 10 µM TBL and 10 µM DBL. 20 µM RNA were used for TBL or DBL only conditions. Subsequently, 0.1 µM tDNA and 0.1 µM dDNA substrates were added to the preassembled ISCro4–RNA complexes at a final concentration of 2 µM protein complex in a total reaction volume of 20 µl recombination buffer. Reactions were incubated at 37 °C for 2 hours. To terminate the reactions, Proteinase K was added, and samples were incubated for 20 min at 50 °C. Reaction mixtures were then diluted 1:10 in RNA denaturation buffer (95% formamide, 25 mM EDTA, 0.15% (w/v) Orange G) and denatured by heating at 95 °C for 15 min. Reaction products were analyzed by denaturing polyacrylamide gel electrophoresis (15% PAGE, 8 M urea), and fluorescent signals were visualized using a Typhoon FLA 9500 biomolecular imager (GE Healthcare Life Sciences). The efficiency of the in vitro recombination reaction was quantified using Image Studio Lite (version 5.2). Three independent biological replicates were performed and the signal intensity of the bands corresponding to cleavage or recombination of the labeled t or dDNA strands (C or R) was divided by the total signal of the corresponding lane (L), and a percentage of converted substrate was calculated as:

$$\%efficiency = \frac{R \text{ or } C}{L} \times 100$$

Cryo-EM analysis

The ISCro4-TBL-DBL-tDNA-dDNA complex was reconstituted in SEC buffer (20 mM Tris-HCl, pH 8.0; 500 mM NaCl, 5 mM $MgCl_2$) by mixing the purified ISCro4 recombinase (final concentration of 1.5 mg/ml), with the in vitro transcribed TBL (111nt) and DBL (79 nt), a 44-bp dDNA and a 38-bp tDNA at a molar ratio of 4:2:1:1:1. The mixture was then incubated at 22°C for 15 min and centrifuged at 20,000 × g and 4°C. Quantifoil Holey Carbon (R1.2/1.3, Au, 200 mesh) grids (SPT Labtech) were glow-discharged in low-pressure air at a 15-mA current for 60 s in a PELCO easiGlow Cryo-EM Glow Discharge system. Using a Vitrobot Mark IV plunger (FEI), 3 µl of the complex sample were applied on the grids, which were blotted with Vitrobot Filter Paper (Grade 595, Electron Microscopy Sciences) for 3 s at 100% humidity and 4°C, plunge frozen in a mix of liquid ethane/propane (37:63 v/v) and stored in liquid nitrogen. Cryo-EM data collection was performed on a FEI Titan Krios G3i microscope (University of Zurich, Switzerland) operated at 300 kV and equipped with a Gatan K3 direct electron detector in super-resolution counting mode. A total of 7,189 movies were recorded at 130,000x magnification, resulting in a super-resolution pixel size of 0.325 Å. Each movie comprised 37 subframes with a total dose of 66.802 $e^-/Å^2$. Data acquisition was performed with EPU Automated Data Acquisition Software for Single Particle Analysis (ThermoFisher Scientific) with three shots per hole at -0.8 µm to -2.6 µm defocus (0.3 µm steps). The collected exposures were processed in CryoSPARC v4.7 (*50*). The 7,189 movies were aligned using *Patch Motion Correction* and their contrast transfer function (CTF) parameters estimated with *Patch CTF Estimation*. Particles were initially selected with *Blob Picker* (minimum particle diameter: 50 Å, maximum particle diameter: 150 Å), extracted with a box size of 400 pixels (Fourier-cropped to 100 pixels), and subjected to *2D Classification* into 200 classes. After selecting 6 of these classes and performing two more rounds of 2D classification with 20 classes, 14 final classes comprising 288,712 particles were selected for *Ab-initio Reconstruction*. The initial volume was refined with *Non-uniform Refinement*, yielding a density map at 5.47 Å resolution (GSFSC). The selected particles were re-extracted with a box size of 400 pixels without Fourier crop and Reference Motion Correction was applied to the micrographs. A final Non-uniform Refinement job using 268,986 particles yielded a map of 2.81 Å resolution. The overall processing workflow is summarized in fig. S6. An initial atomic model of the ISCro4 complex was generated based on the IS621 synaptic complex structure (PDB ID: 8WT6) by substituting differing amino acids and nucleotides. The model was manually rebuilt in Coot (*51*) and refined using real space refinement in Phenix (*52*) against the corresponding unsharpened cryo-EM map. Secondary structure restraints were initially applied but released in the final refinement cycle. The model quality was validated using MolProbity (*53*). Reconstruction and refinement statistics are provided in table S4. Figures of the map and model were prepared using UCSF ChimeraX (*54*).

Direct amplicon sequencing analysis

Inversion and insertion samples were evaluated using a direct amplicon sequencing approach: Genomic DNA from samples were isolated by the DNeasy Blood & Tissue Kit (Qiagen, 69504) following the manufacturer's protocol. Locus-specific primers were used to generate targeted amplicons for deep sequencing. Donor plasmids used to quantify chromosomal insertions were engineered to include locus-specific primer binding sites, resulting in amplicons of identical size for edited and unedited samples. The corresponding sequences are listed in table S1. The first PCR step used primers containing Illumina adapter overhangs; the sequences of all primers used are listed in table S1. A second PCR step was carried out to introduce sample-specific p5–p7 barcodes. PCR products were purified using the NucleoSpin Gel and PCR Clean-up kit (Macherey-Nagel, 740610.20) following the manufacturer's protocol. The final library pool was quantified using a Qubit 3.0 fluorometer (Invitrogen), and sequencing was performed on an Illumina MiSeq platform (150 bp, paired-end reads). Amplicon sequences were analyzed using custom Python scripts (https://github.com/Schwank-Lab/ISCro4_Pelea_2026) (*46*).

Fragment amplicon sequencing analysis

For *CNBP* deletions, a modified version of the GUIDE-seq protocol (*55*) was used as described below. Genomic DNA was purified three days post-transfection according to the Puregene DNA Purification protocol (Gentra systems). 200 ng of genomic DNA was sheared with the LE220-plus Focused-ultrasonicator (Covaris) to 550 bp in average in microTUBE-130 AFA tubes, followed with end-repair and adapter (containing unique molecular identifiers (UMIs)) ligation and bead cleanup as described previously. Next, linear PCR was performed with the outer primers (table S1) for 45 cycles using 0.15 µl Platinum Taq polymerase (Thermo Fisher Scientific), 0.75 µl of 0.5 M TMAC, 0.75 µl of 1 µM outer primer in a 15 µl reaction, according to the manufacturer's instructions. After AMPure XP 0.9× bead cleanup, the first step and second step PCR reactions were performed as described previously (except site specific i7 primer was used, for details see table S1). The final library was size selected on a 2% agarose gel for fragments between 150 bp and 400 bp, gel extracted with the NucleoSpin Gel and PCR Clean-up kit (Macherey-Nagel) quantified with Qubit HS dsDNA Assay (Thermo Fisher Scientific) and KAPA Library Quantification Kit (Roche) and sequenced on a MiSeq (Illumina). Sequencing data were processed using a custom Snakemake (*56*) pipeline. Raw reads were demultiplexed, deduplicated using 8 bp UMIs combined with sequence context, adapter-trimmed with Cutadapt (*57*), and classified as wild-type, edited, or unintended based on exact sequence matching (https://github.com/Schwank-Lab/ISCro4_Pelea_2026) (*46*).

CUT&RUN assay

Cleavage Under Targets and Release Using Nuclease (CUT&RUN) was performed as previously described (*58*). Briefly, HEK293T cells were transfected with 3xFLAG-tagged ISCro4 as well as TBL and DBL plasmids using the TurboFect transfection reagent (Thermo Fisher Scientific, R0531). Cells were harvested 9 hours post-transfection and were lightly fixed in 0.1% PFA followed by nuclei extraction and immobilization on concanavalin A–coated magnetic beads. Samples were incubated with anti-FLAG tag antibody (Cell Signaling Technologies 87537S) targeting the 3xFLAG-tagged ISCro4, followed by incubation with a protein A/G-Micrococcal Nuclease (pAG-MNase) fusion protein. Antibody-associated DNA fragments were released by pAG-MNase digestion upon adding calcium and incubating for 30 min on ice. Samples were then de-crosslinked and DNA fragments were purified by solid-phase reversible immobilization purification. Sequencing libraries were prepared using the NEBNext Ultra II kit (New England Biolabs E7103S) following the manufacturers' instructions. Next-generation sequencing of libraries was performed using the Illumina NovaSeq X platform as 2x150bp paired-end reads by the Functional Genomics Center Zurich (ETH Zürich, Universität Zürich) at a minimum read depth of 25E6 reads.

CUT&RUN data processing and analysis

Raw NGS reads were trimmed for the Illumina adapter sequences using *cutadapt* (version 5.0) followed by alignment against a custom genome index comprising the human genome *GRCh38* (UCSC) main chromosomes and transfected ISCRo4 plasmid sequences using *bowtie2* (version 2.5.4) and *samtools* (version 1.21); thereafter, the BAM datafiles were deduplicated using *samtools* and converted to counts-per-million (CPM) normalized coverage using *deepTools bamCoverage* (version 3.5.5) (*57*, *59*–*62*). ISCro4 binding sites were determined using a directed search strategy (https://github.com/cornlab/bits) (*63*). First, all possible 0-4 mismatch human genomic *AAVS1*-targeting guide RNA sites were identified using *Cas-OFFinder* (version 2.4.1) (*64*). For each site and sample, the region total CPM was calculated independently for the 500bp regions upstream and downstream of the candidate bridge RNA target site. Thereafter the following values were computed by comparing a ISCro4-transfected sample and the untransfected background control: background-subtracted region total CPM, log2 fold-enrichment (log2FE) of sample over background. Regions were then filtered for those passing the following criteria for both upstream and downstream regions: background-subtracted region total CPM ≥ 1, log2FE ≥ 1, and relative difference between the upstream and downstream background-subtracted total CPM ≤ 0.20. Additional R packages used include: *tidyverse* (version 2.0.0) (*65*), *GenomicRanges* (version 1.60.0) (*66*), *BSgenome.Hsapiens.UCSC.hg38* (version 1.4.5) (10.18129/B9.bioc.BSgenome.Hsapiens.UCSC.hg38) and *vroom* (1.6.6) (https://vroom.r-lib.org). Each site was then scored by multiplying the log2FE and background-subtracted

CPM (log2FE × background-subtracted CPM) and summing these values for both the upstream and downstream regions. Sites overlapping previously reported genomic blacklist regions (ENCODE project ENCFF356LFX; *Nordin et al.* (2023) CUT&RUN blacklist) were removed (*67*–*70*). Final hits (tables S5 and S6) were determined by filtering for sites with both score ≥ 30 and having the "CT" motif at positions 8-9 in the target sequence; nonspecific high signal regions were removed by filtering for score < 10000. Hits were summarized using a modified "draw_blender_fig.py" from *BLENDER* (https://github.com/cornlab/blender) and *ggseqlogo* (version 0.2) (*71*, *72*).

Genome-wide distribution of ISCro4-targetable sites

All possible 14-nucleotide sequences containing a CT dinucleotide core at positions 8-9 (16,777,216 total sequences) were generated computationally and aligned to the human reference genome (GRCh38/hg38) using Bowtie2 v2.5.1 (*59*) with parameters requiring perfect matches (-k 150 -N 0–score-min L,0,0–no-1mm-upfront–end-to-end). Sequences were classified by genomic occurrence as not found (0 matches), unique (1 match), dual (2 matches), or ≥3 matches. Unique target sites were further annotated against protein-coding genes from ENSEMBL release 114 (*73*), including 1 kb flanking regions, using bedtools v2.30.0 (*74*). Gene features were classified using canonical transcripts, with specific annotation of first exons and first introns. Pathogenic gene analysis utilized ClinVar variant data (GRCh38 assembly, downloaded July 30, 2025) (*75*), filtered for single-gene variants with "Pathogenic" clinical significance, yielding 177,014 pathogenic variants across 4,786 genes from the 20,120 protein-coding genes analyzed. For additional information, see (https://github.com/Schwank-Lab/ISCro4_Pelea_2026) (*46*).

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SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.adz1884
Figs. S1 to S18
Tables S1 to S6
MDAR Reproducibility Checklist
Data S1

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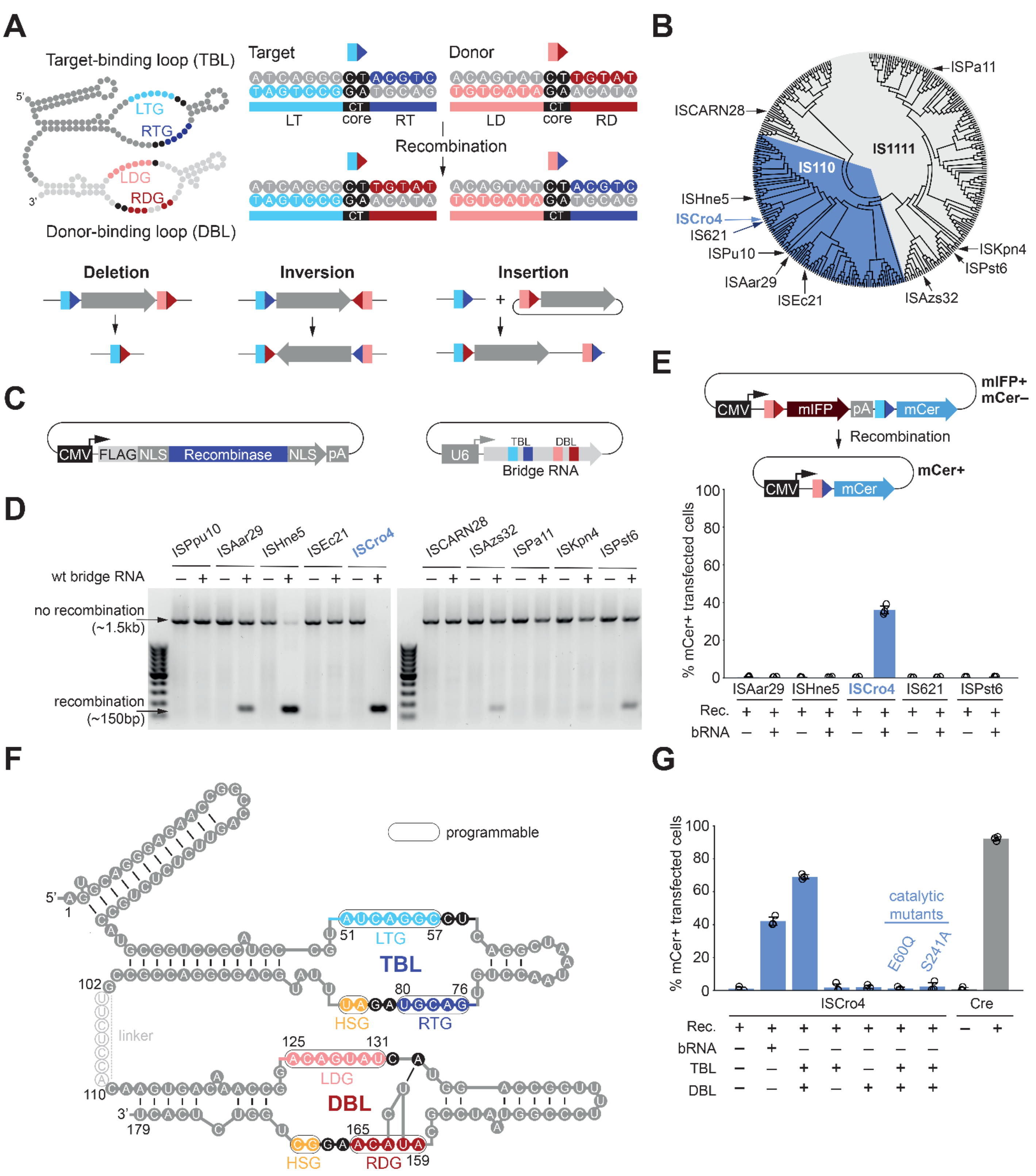

A
Target-binding loop (TBL)
LTG
RTG
LDG
RDG
Donor-binding loop (DBL)
Target
Donor
LT
core
RT
LD
RD
Recombination
Deletion
Inversion
Insertion
B
ISPa11
ISCARN28
IS1111
IS110
ISHne5
ISCro4
IS621
ISPu10
ISAar29
ISEc21
ISAzs32
ISKpn4
ISPst6
C
CMV
FLAG
NLS
Recombinase
NLS
pA
U6
TBL
DBL
Bridge RNA
D
ISPpu10
ISAar29
ISHne5
ISEc21
ISCro4
ISCARN28
ISAzs32
ISPa11
ISKpn4
ISPst6
wt bridge RNA
no recombination (~1.5kb)
recombination (~150bp)
E
CMV
mIFP
pA
mCer
mIFP+ mCer–
Recombination
mCer+
% mCer+ transfected cells
ISAar29
ISHne5
ISCro4
IS621
ISPst6
Rec.
bRNA
F
programmable
5'
1
51
LTG
57
TBL
102
80
76
HSG
RTG
linker
125
131
LDG
DBL
110
3'
179
165
HSG
RDG
159
G
% mCer+ transfected cells
catalytic mutants
E60Q
S241A
ISCro4
Cre
Rec.
bRNA
TBL
DBL

Fig. 1. In cellulo screening of bridge recombinases. (**A**) Bridge recombinases use a bridge RNA to mediate recombination between 14-bp target and donor sites, each defined by the presence of a core dinucleotide. Bridge RNAs comprise a target-binding loop (TBL) and a donor-binding loop (DBL). The TBL contains a left-target guide (LTG) and right-target guide (RTG) segments, which base pair with the bottom strand of the left target (LT) and top strand of the right target (RT), respectively. The DBL includes left-donor guide (LDG) and right-donor guide (RDG) segments, which base pair with the bottom strand of the left donor (LD) and top strand of the right donor (RD). Depending on the relative orientation of target and donor sites, bridge recombinases mediate RNA-programmable deletions, inversions, or insertions. (**B**) Neighbor-joining phylogenetic tree constructed for 332 IS110 family recombinases listed in the ISFinder database (*37*). IS110 family is divided into the IS110 and IS1111 subgroups. Orthologs selected for screening are labeled. (**C**) Schematic representation of the expression constructs for bridge RNA and recombinase screening. (**D**) Nested PCR analysis of candidate recombinases from the IS110 (left) and IS1111 (right) subgroups. Recombination-mediated deletion of a ~1.35 kb fragment results in the generation of a 150-bp PCR product. (**E**) (Top) Recombination fluorescent reporter design. Recombination results in the deletion of mIFP expression cassette, resulting in the activation of mCerulean fluorescent protein (mCer) expression. Bottom: Activity of selected recombinases in HEK293T, assessed by measuring the fraction of mCer-positive cells within the transfected cell population (denoted as % mCer+ transfected cells), i.e., cells positive for mIFP or mCer expression. Bar graph shows the mean ± standard deviation (SD) from three independent biological replicates (n = 3). (**F**) Schematic representation for the split bridge RNA design. The TBL sequence includes nucleotides 1-102 from the full bridge RNA sequence, including a 5′-terminal hairpin (*33*). The DBL includes bridge RNA nucleotides 110-179. The deleted linker sequence is indicated with a dashed line. (**G**) Fluorescent reporter assay of recombination activity of ISCro4 programmed with the split-bridge TBL and DBL RNAs, performed and analyzed as in (E). Cre recombinase (used with a similar reporter plasmid) served as a positive control. Bar graph shows the mean ± standard deviation (SD) from three independent biological replicates (n = 3).

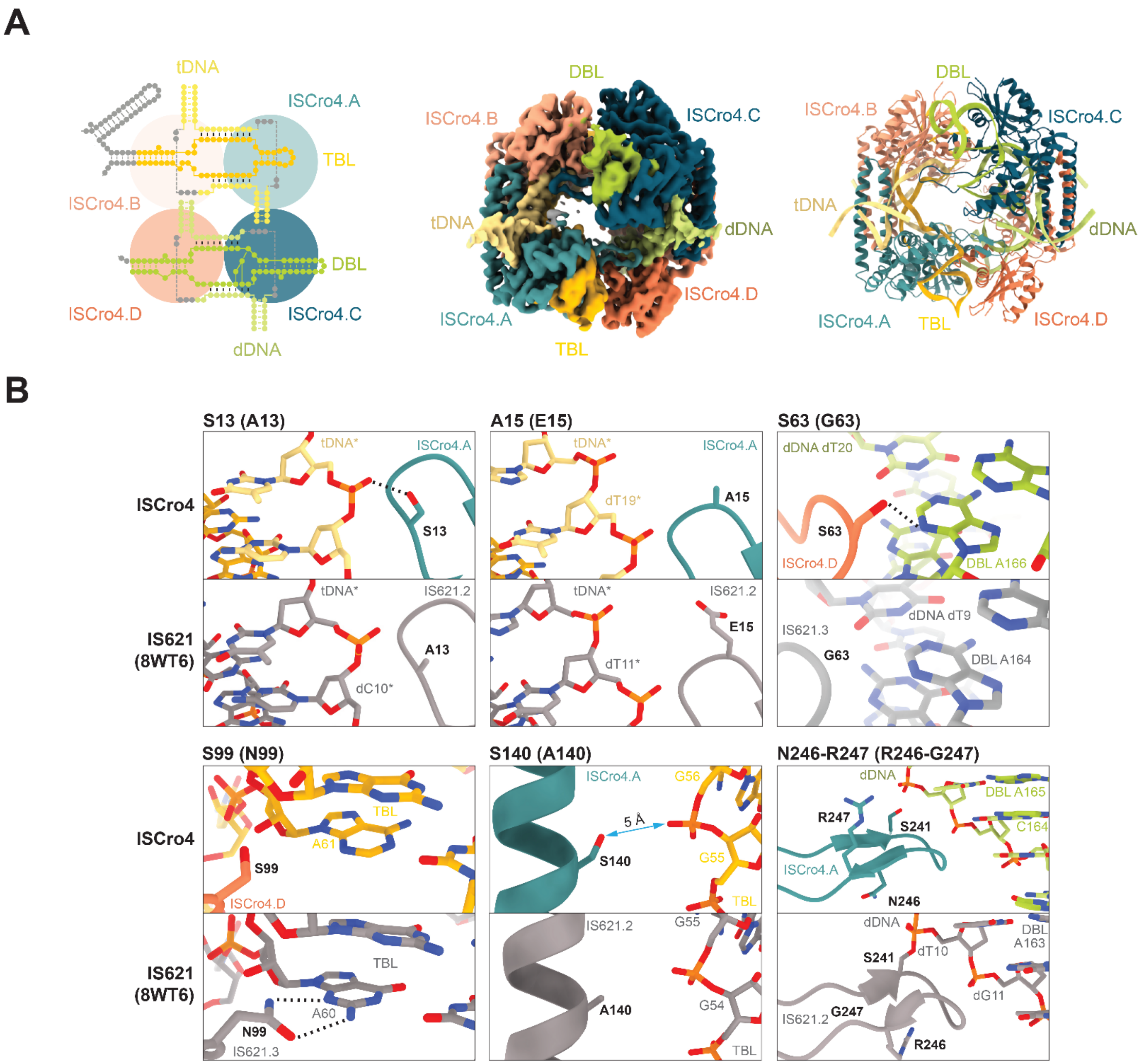
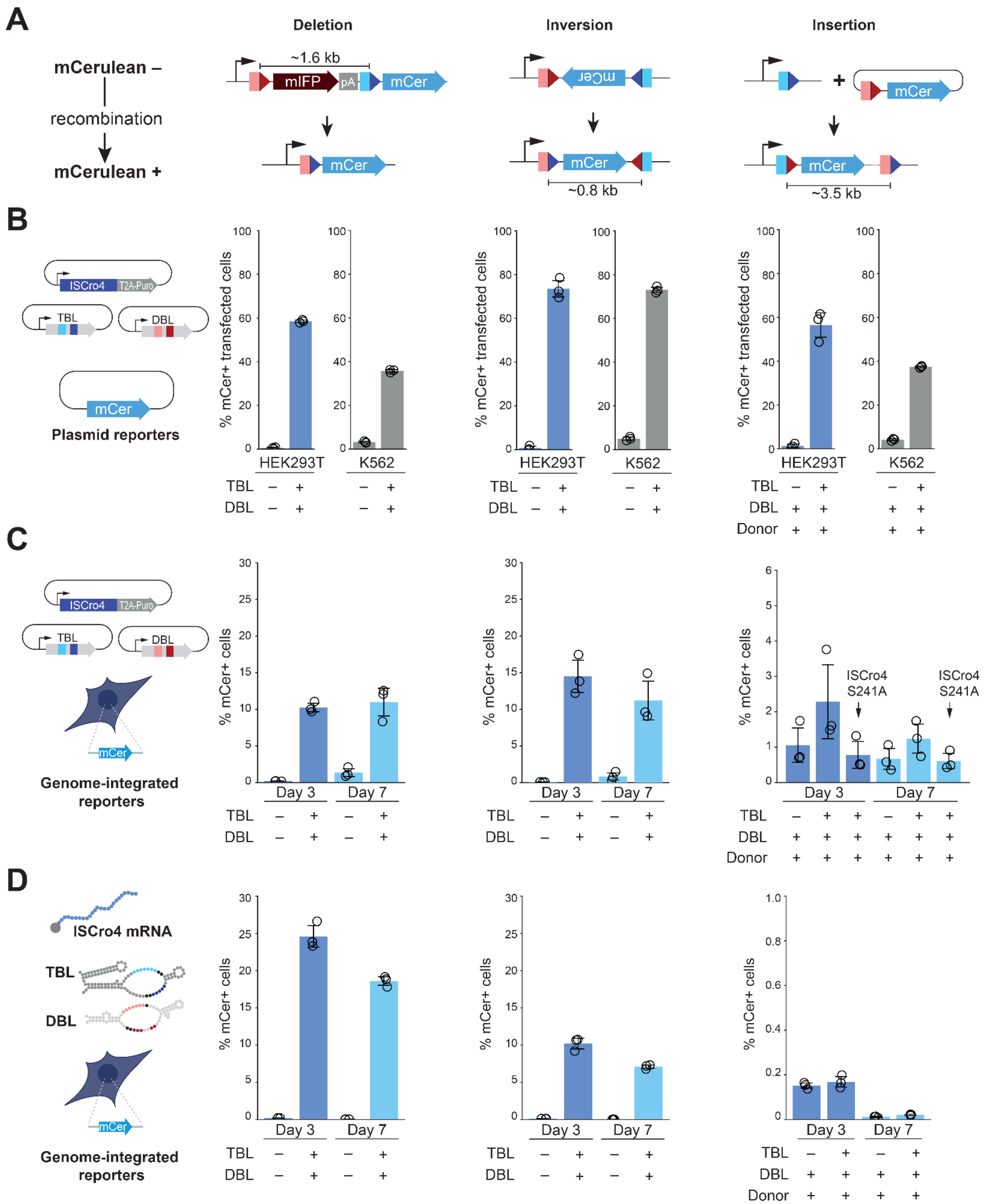


Fig. 2. **Structural insights into the enhanced activity of ISCro4.** (A) (Left) Schematic representation of the pseudo-tetrameric ISCro4-TBL-DBL-tDNA-dDNA synaptic complex. Middle: Experimental cryo-EM density map of the complex at a resolution of 2.81 Å. ISCro4 subunits are colored in blue and orange; TBL and tDNA in shades of yellow; DBL and dDNA in shades of green. (Right) Ribbon representation of the complex, shown in the same orientation. (B) Representative close-up views of amino acids that differ between ISCro4 (colored blue or orange) and IS621 (PDB 8WT6, shown in grey) and are involved in interactions with bound guide RNAs or DNAs. Hydrogen bonding interactions and interatomic distances are shown as blue dashed and solid lines, respectively. An asterisk indicates that the bottom strand of the DNA is shown.



A
Deletion
Inversion
Insertion
mCerulean –
recombination
mCerulean +
~1.6 kb
mIFP
pA
mCer
~0.8 kb
~3.5 kb
B
ISCro4
T2A-Puro
TBL
DBL
Plasmid reporters
% mCer+ transfected cells
HEK293T
K562
Donor
C
Genome-integrated reporters
% mCer+ cells
Day 3
Day 7
ISCro4 S241A
D
ISCro4 mRNA

Fig. 3. ISCro4 mediates genomic deletions, inversions, and insertions. (**A**) (Top) Schematic representation of mCerulean reporter constructs used to test ISCro4-mediated deletions, inversions, and insertions. (Bottom) Positive recombination events result in activation of mCerulean (mCer) protein expression. (**B**) Recombination of deletion, inversion and insertion reporters carried on episomal plasmids in HEK293T and K562 cells, quantified as fraction of mCer-positive cells in the transfected cell population (% $mCer^+$ transfected cells) by flow cytometry. Bar graphs show the mean ± standard deviation (SD) from three independent biological replicates (n = 3). (**C**) Recombination of deletion, inversion and insertion reporters stably integrated into the genome of HEK293T, quantified as fraction of mCer-positive cells by flow cytometry. ISCro4 and TBL/DBL RNAs were expressed from transfected plasmids. Bar graphs show the mean ± standard deviation (SD) from three independent biological replicates (n = 3). Flow-cytometry analysis was done on days 3 and 7 post-transfection. Puromycin was applied 24 hours post transfection for 48 to select for cells harboring the ISCro4-T2A-puro expression plasmid. For insertions, catalytically inactive ISCro4 mutant S241A served as negative control to account for non-specific mCer activation. (**D**) Recombination of deletion, inversion and insertion reporters stably integrated into the genome of HEK293T, quantified as fraction of mCer-positive cells by flow cytometry. Cells were electroporated with ISCro4-encoding mRNA, synthetic TBL/DBL RNAs and, for insertions, with donor DNA plasmid. Bar graphs show the mean ± standard deviation (SD) from three independent biological replicates (n = 3). Flow-cytometry analysis was done on days 3 and 7 post-transfection.

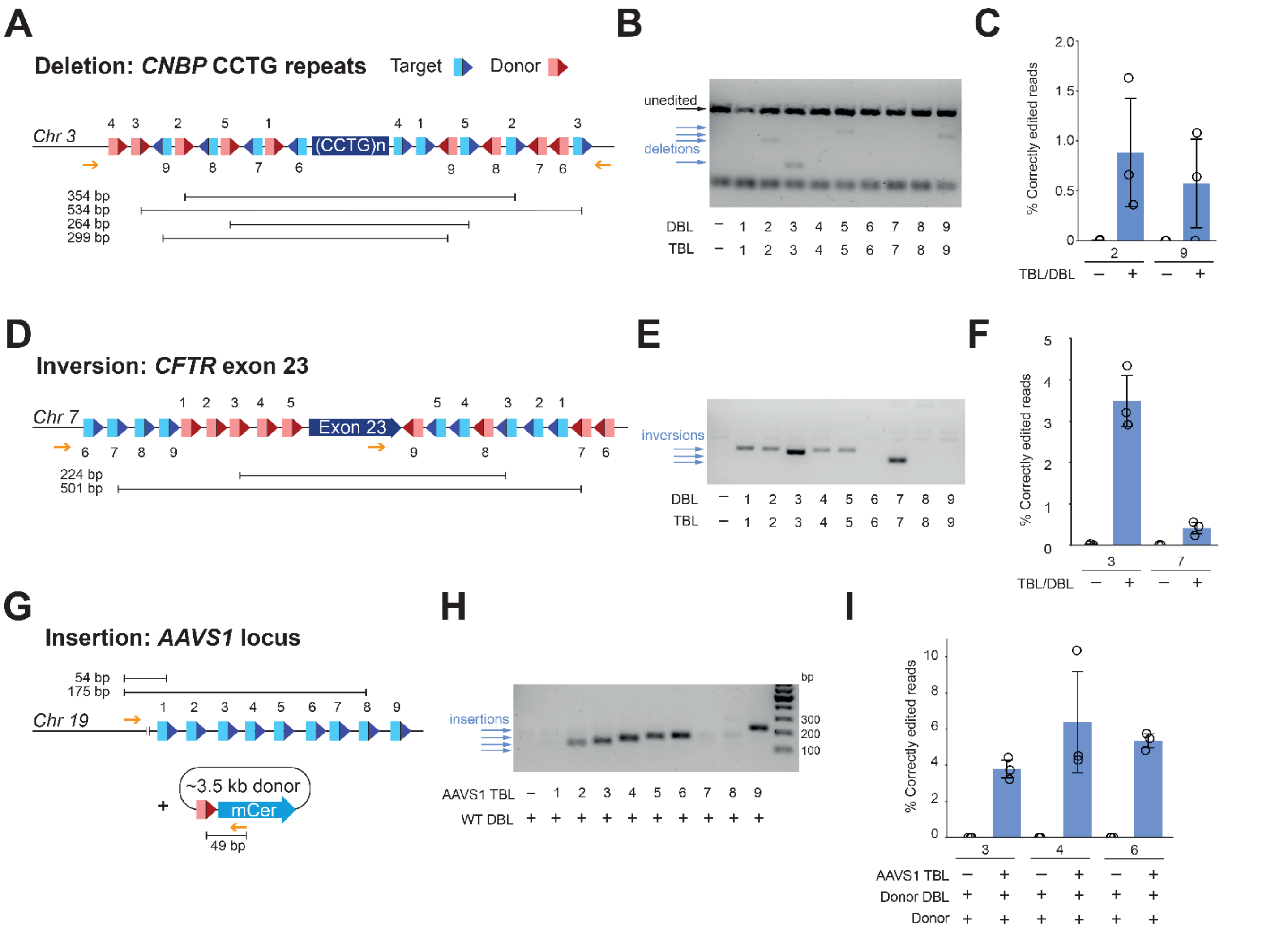


Fig. 4. ISCro4 enables editing of endogenous genomic loci. (**A**) Schematic representation of ISCro4-mediated excision of CCTG tetranucleotide repeats from the *CNBP* locus on chromosome 3. Nine target–donor pairs in the same orientation were selected from sequences flanking the CCTG repeats. (**B**) Nested PCR analysis of recombination outcomes for all nine TBL-DBL pairs in HEK293T cells. (**C**) Next-generation sequencing (NGS) quantification of the excision efficiency for two functional TBL–DBL pairs, expressed as the fraction of reads containing a correctly recombined donor-target junction sequence within all mapped reads. Bar graph shows the mean ± standard deviation (SD) from three independent biological replicates (*n* = 3). (**D**) Schematic representation of ISCro4-mediated inversion of exon 23 in the *CFTR* gene on chromosome 7. Nine target–donor pairs in opposite orientation were selected from sequences flanking the exon. (**E**) Nested PCR analysis of recombination outcomes for all seven TBL-DBL pairs. PCR amplification occurs only in the presence of successful inversion events. (**F**) NGS quantification of the inversion efficiency for two functional TBL–DBL pairs, expressed as the fraction of inversion-containing reads within all mapped reads. Bar graph shows the mean ± standard deviation (SD) from three independent biological replicates (*n* = 3). (**G**) Schematic representation of insertion of a ~3.5 kb donor plasmid at the *AAVS1* safe-harbor locus on chromosome 19. Nine potential target sites were selected for TBL guide design and used together with a DBL guide complementary to native ISCro4 donor sequence within the donor plasmid. (**H**) Nested PCR analysis of integration events at the *AAVS1* locus, indicated by amplification products ranging from 103 bp to 224 bp (**I**) NGS quantification of the insertion efficiency for three functional TBL guides, expressed as the fraction of reads containing a correctly recombined donor-target junction sequence within all mapped reads. Bar graph shows the mean ± standard deviation (SD) from three independent biological replicates (*n* = 3).

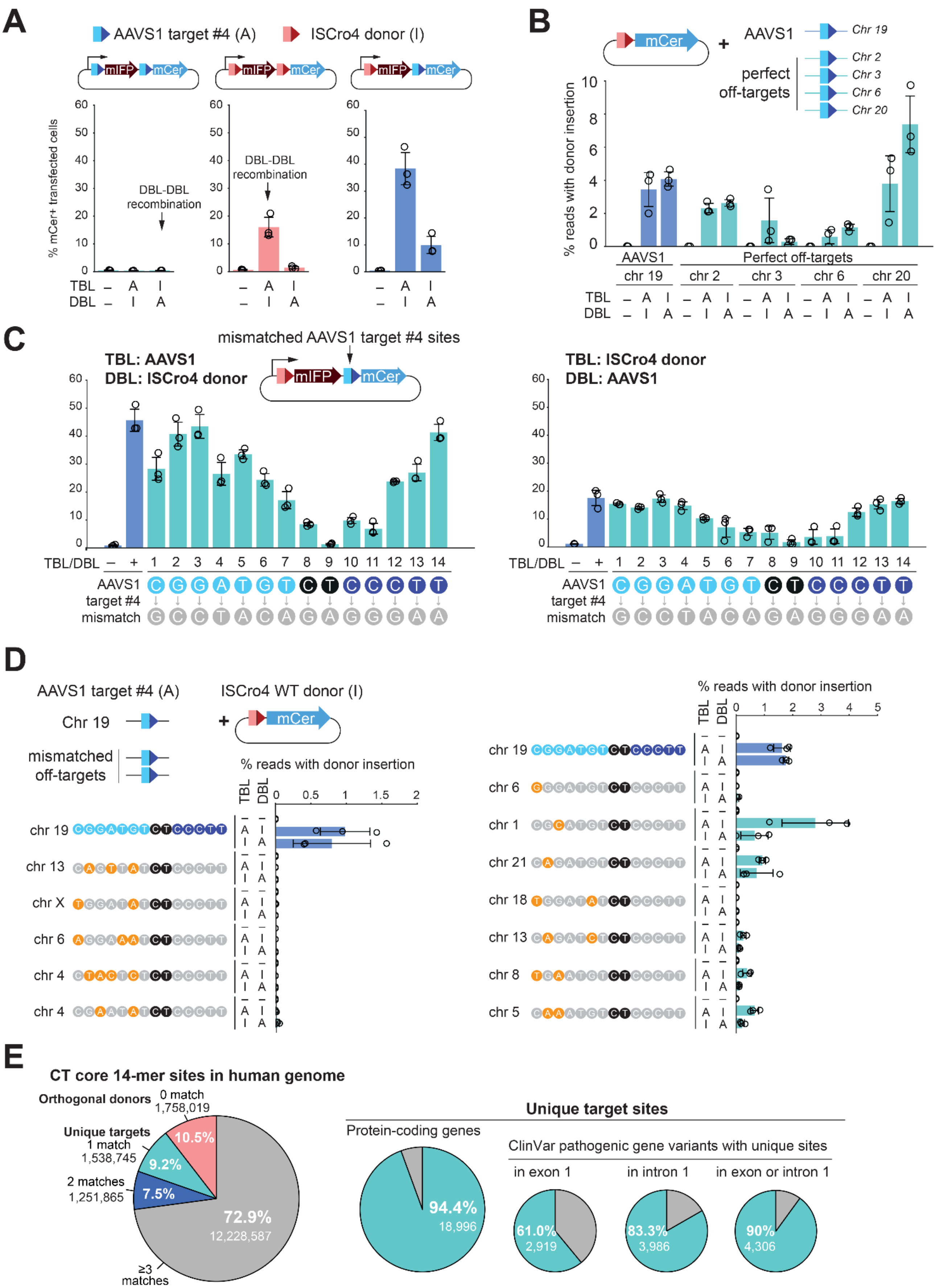

A
AAVS1 target #4 (A)
ISCro4 donor (I)
mIFP
mCer
% mCer+ transfected cells
DBL-DBL recombination
TBL
DBL
B
mCer
AAVS1
Chr 19
perfect off-targets
Chr 2
Chr 3
Chr 6
Chr 20
% reads with donor insertion
AAVS1
Perfect off-targets
chr 19
chr 2
chr 3
chr 6
chr 20
C
mismatched AAVS1 target #4 sites
TBL: AAVS1
DBL: ISCro4 donor
TBL: ISCro4 donor
DBL: AAVS1
TBL/DBL
AAVS1 target #4
mismatch
D
AAVS1 target #4 (A)
Chr 19
mismatched off-targets
ISCro4 WT donor (I)
% reads with donor insertion
E
CT core 14-mer sites in human genome
Orthogonal donors
0 match
1,758,019
Unique targets
1 match
1,538,745
2 matches
1,251,865
≥3 matches
10.5%
9.2%
7.5%
72.9%
12,228,587
Unique target sites
Protein-coding genes
94.4%
18,996
ClinVar pathogenic gene variants with unique sites
in exon 1
in intron 1
in exon or intron 1
61.0%
2,919
83.3%
3,986
90%
4,306

Fig. 5. Addressing ISCro4 specificity constraints. (**A**) Plasmid-based recombination assays assessing TBL-to-TBL and DBL-to-DBL recombination at the AAVS1 target #4 (A) and ISCro4 native donor (I) sites in HEK293T cells. Recombination efficiency was quantified by flow cytometry as the fraction of mCer-positive transfected cells (% $mCer^+$ transfected cells). Bar graph represents the mean ± standard deviation (SD) from three independent biological replicates (n = 3). (**B**) NGS quantification of donor plasmid integration at the *AAVS1* locus and at four perfect off-target sites identical to the AAVS1 target #4 sequence in HEK293T cells. Integration efficiencies were expressed as the percentage fraction of reads containing a correctly recombined donor-target junction sequence within all mapped reads. (**C**) Plasmid-based recombination assays evaluating the mismatch tolerance of TBL and DBL guides targeting the AAVS1 target #4 site in HEK293T cells. Single-nucleotide mismatches were introduced in the plasmids, while the TBL and DBL guide sequences remained unchanged. Recombination activity was quantified by flow cytometry as the fraction of mCer-positive cells within the transfected cell population (% $mCer^+$ transfected cells). Bar graph shows mean ± standard deviation (SD) from three biological replicates (n = 3). (**D**) NGS quantification of donor plasmid integration at the *AAVS1* locus and a subset of mismatched off-target binding sites identified by CUT&RUN analysis. Left and right panels represent separately performed experiments using distinct donor plasmids containing primer binding sites to quantify integration at the indicated off-target site by NGS. Insertion efficiency is expressed as the percentage fraction of reads containing a correctly recombined target-donor junction sequence within all mapped reads. Bar graph shows mean ± standard deviation (SD) from three biological replicates (n = 3). (**E**) (Left) Classification of all possible 14-nucleotide sequences containing a CT dinucleotide core at positions 8-9 (16,777,216 total sequences) in the human reference genome (GRCh38/hg38) based on their occurrence. (Middle) Proportion of human protein-coding genes (n = 20,120) that contain at least one unique CT-core 14-mer site within the gene body or 1 kb upstream and downstream flanking regions. (Right) Proportion of ClinVar pathogenic genes (n = 4,786) that contain unique 14-CT-core mer sites in exon 1 or intron 1, or either exon 1 and/or intron 1. Teal-colored sections indicate genes with unique sites; gray sections indicate genes lacking unique sites.

Programmable genome editing in human cells using RNA-guided bridge recombinases

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